

The argument structure of Turkish verbal reflexives*

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Abstract: We leverage the rich diagnostic possibilities available in Turkish to probe the representation of verbal reflexivity in this language. We demonstrate that the sole syntactic argument of Turkish verbal reflexives, the surface subject, shows evidence of a VP-internal origin, but also passes diagnostics otherwise only satisfied by deep subjects. We argue that this mixed behavior reflects a movement derivation: a reflexive flavor of the agent-introducing head Voice attracts the sole argument from a low base position to its specifier and derives the reflexive interpretation through VoiceP-internal binding of a free variable. Phrasal movement — a *bona fide* narrow-syntactic operation — can then be crucially implicated in the derivation of verbal reflexivity.

Keywords: verbal reflexives; argument structure; A-movement; Voice; Turkish

1 Introduction

We discuss the syntax of Turkish verbal reflexives (TVRs), (1). The verb in this example supports a passive reading, according to which *Leyla* was washed by a disjoint agent; but also a reflexive reading, whereby *Leyla* is both the agent and the theme of the washing. Past studies of TVRs (Key, 2023, 2024) leave open the provenance of the surface subject of these verbs, our main focus here.

- (1) Leyla soğuk su-yla yıka-**n**-dı-Ø.
Leyla.NOM cold water-with wash-NACT-PST-3SG
‘Leyla washed herself with cold water.’ OR ‘Leyla was washed with cold water.’

The general descriptive profile of these verbs is common cross-linguistically (e.g. Geniušienė, 1987; Kemmer, 1993; Reuland, 2018; Alexiadou and Schäfer, 2013): (1) involves a single overt argument, the surface subject; and a particular kind of voice morphology, also borne by passives in the language, with verbal reflexivity never being possible with morphologically active verbs, (2).

- (2) Leyla soğuk su-yla yıka-dı-Ø.
Leyla.NOM cold water-with wash-PST-3SG
‘Leyla washed (something) with cold water’; NOT ‘L was washed’ or ‘L washed herself’

As in other languages, the reflexivization strategy in (1) is restricted in two ways not typical of pronominal reflexives in the language. TVRs are lexically restricted: not every Root forms a verbal reflexive. They are also thematically restricted in the sense that verbs incompatible with Agents (e.g. experiencer predicates) cannot form verbal reflexives (for overview see, Reuland, 2018).

A first question raised by verbal reflexives concerns the extent to which they involve pronominal anaphora of some kind: to what extent is (1) similar to (3), where the source of reflexivity is a

pronominal reflexive co-indexed with a distinct nominal? A second question concerns what happens if it turns out that the two strategies are dissimilar: what does it mean to be a verbal reflexive? Both questions require precision on how reflexivity is encoded in the reflexive parse of (1), that is, what guarantees that *Leyla* is ultimately understood both as the agent and as the theme of the washing.

- (3) Leyla **kendin-i** soğuk su-yla yıka-dı-Ø.
 Leyla.NOM self-ACC cold water-with wash-PST-3SG
 ‘Leyla washed herself with cold water.’

Different approaches maintain that the phrasal syntax plays a central role in deriving reflexive interpretations not just in (3)—where the centrality of syntax to pronominal reflexivity is uncontroversial (see e.g. Chomsky, 1973, 1981; Reuland, 2011; Reinhart, 1983; Reinhart and Reuland, 1993; Büring, 2005; Safir, 2004)—but also in (1). Many syntactic approaches employ *movement* as an operation crucially implicated in the derivation of reflexivity, as discussed below.

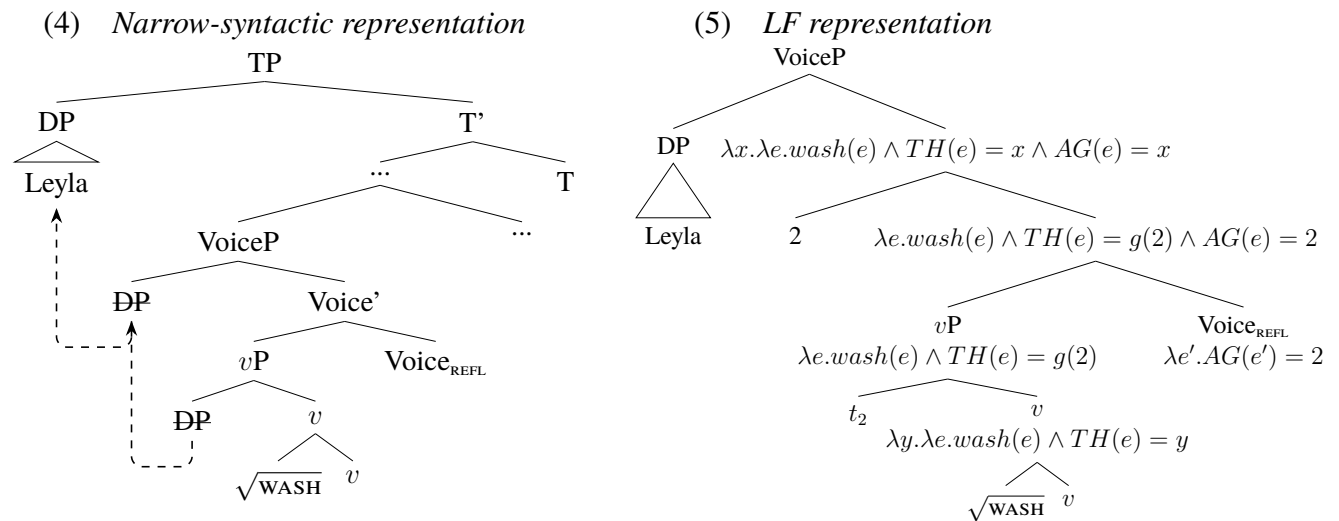
We argue that TVRs instantiate a case where a movement-based derivation of verbal reflexivity is evidenced with particular clarity. Turkish makes available a cluster of phenomena that help diagnose with unusual precision the derivational history of the surface subject of verbs like the one in (1). We first show (in §2) that such verbs are syntactically intransitive: in (1), *Leyla* is the only syntactic argument in the structure. We then show (in §3) that the surface subject originates as a deep object; TVRs thus have the basic configurational syntax of unaccusatives/passives, with the deep object ultimately raising to become the grammatical subject. Interestingly, however, we demonstrate in §4 that this argument transits through an intermediate derived subject position in the specifier of VoiceP; crucially, this intermediate step is only evidenced for reflexives, and not for passives or unaccusatives. We argue that these observations, which arise from diagnostics sensitive to syntactic position and not merely to thematic roles, support movement as a syntactic route to reflexivization in natural language.

At the same time, the facts demand a particular view of movement-based identification. Turkish shows that verbal and pronominal reflexives pattern in crucially different ways, such that the movement-based derivation cannot be extended to pronominal reflexives; there is no anaphor to bind in the syntax of TVRs, so the derivation of the reflexive reading cannot involve a pronominal element *per se*; and it can be shown that the movement step responsible for reflexivization is not to the grammatical subject position, but to a lower VoiceP-internal derived position.

Existing movement-approaches come in three variants, none of which suffices for our data. A standard analysis of Romance clitic reflexives involves movement of a deep object across the clitic subject to surface subject position in a way that ensures Condition A-compliant syntactic binding and produces thematic role identification (e.g., Marantz 1984; Kayne 1988; Pesetsky 1995; McGinnis

2004; Sportiche 2005). Since we show there is no pronoun involved in the structure of (1), this type of approach is not available to us without modification. The Romance approach has been extended, in modified form, to pronoun-less reflexives similar to TVRs (see e.g. Embick 2004b; Spathas et al. 2015; Kastner 2017)–but these analyses only posit movement to the grammatical subject position, while our data crucially suggest that it is an intermediate landing site that is crucial for reflexivization. Closest in spirit to the analysis we propose for Turkish is Hornstein (1999, 2001); Boeckx et al. (2008), where reflexivization of different kinds is treated as movement between thematic positions. Our analysis is broadly reminiscent of this general approach, but ultimately differs in important ways in both implementation and scope. We elaborate after presenting our analysis in §5.

Our analysis is previewed in (4) and (5). We argue that the agent-introducing head Voice comes in a reflexive variant, Voice_{REFL} (Labelle 2008; McGinnis 2022; Paparounas 2023; Kallulli and Roberts 2025; Oikonomou and Alexiadou 2022; Raghotham 2022; cp. Ahn 2015; Kratzer 2009; Baker 2022; Paparounas and Akkuş 2024 for related ideas in *pronominal* reflexivity). Voice_{REFL} does not introduce an externally merged specifier, but it triggers movement of a lower element – typically, the complement of the complex head *v* – to its specifier. (We adopt acategorial Roots head-adjoined to categorizers; alternatives would leave the analysis unchanged). At LF, Voice_{REFL} introduces an Agent role saturated by a free variable; movement of a lower element to the specifier of Voice results in VoiceP-internal binding of this free variable, effectively reflexivizing the Agent role. (Throughout, we notate movement to Spec,TP only when relevant; since the denotations of Asp and T are not crucial, our LF trees are limited to the VoiceP).



The bigger-picture upshot of our discussion concerns the role of syntax: since movement is a *bona fide* narrow-syntactic operation, TVRs must be constructed syntactically. All things being equal,

this result is compatible both with analyses of verbal reflexives couched in theories that eschew pre-syntactic word-formation across the board (Kastner, 2017; Alexiadou and Schäfer, 2013; Wood, 2014; Paparounas, 2023; McGinnis, 2022; Spathas et al., 2015); and with theories that countenance syntactic reflexivization as a parametric option alongside a lexical strategy (Reinhart and Siloni 2004, 2005; Siloni 2012; Papangeli 2004; cp. Lidz 1996, 2001a,b; Bresnan 1982; Wehrli 1986; Grimshaw 1982; Bouchard 1982; Chierchia 2004; and see Marelj and Reuland 2016 for a reduction of the parameter). We state our analysis in the vocabulary of the former type of theory; at a minimum, our analysis can be seen as giving substance, for a language with the properties of Turkish, to the notion of *syntactic bundling* invoked in Reinhart and Siloni (2004, 2005).

We do not attempt to adjudicate further between the two classes of theories, reserving broader architectural comparisons for a different occasion. We note, however, that the fact that TVRs demand a syntactic analysis may ultimately prove informative as to the criteria that the latter type of theory uses to classify a given reflexivization strategy as ‘lexical’ or ‘syntactic’ in the first place. For instance, lexical restrictedness is taken to be a hallmark of ‘lexical’ reflexivization in Reinhart and Siloni (2005); but TVRs are built syntactically despite the fact that not every Root in Turkish can result in a TVR. From the perspective of a syntactic theory of word-formation, this is not surprising: nothing in a theory combining Roots with verbal functional structure demands that every combination of the two be licit (see e.g. Marantz, 1997; Alexiadou and Schäfer, 2013; Alexiadou and Doron, 2012; Alexiadou and Anagnostopoulou, 2004; Alexiadou et al., 2006; Schäfer, 2008; Pytkänen, 2008; Alexiadou and Doron, 2012; Borer, 2013; Alexiadou et al., 2015; Wood, 2015; Kastner, 2017; Jo, 2019).

Note, finally that the notion of lexical restrictedness itself must be made precise. Like verbal reflexives in other languages, TVRs have been described as being formed only from a set of Roots which is *i)* small in cardinality and *ii)* limited to predicates denoting different kinds of self-action, such as grooming verbs (see e.g. Kornfilt 1997:140; Kemmer 1993; Geniušienė 1987). Our investigation has revealed that, for Turkish at least, the class of Roots that form verbal reflexives is in fact neither small nor clearly lexicosemantically circumscribed, even on a generous conception of the notion of self-orientedness. Recent work on the language reaches the same conclusion (see Arslan 2022 for TVRs and Atlamaz and Öztürk 2023 for Turkish verbal reciprocals; other languages have been noted to behave similarly, e.g. Dutch in Reuland 2011, 199). There are hundreds of TVR-forming Roots, of which many if not most are not ‘inherently’ reflexive in any obvious way (e.g. *bağla-n* ‘attach’, *silkele-n* ‘dust off [figuratively]’, *salla-n* ‘swing (e.g., in a swing)’, *sığ-in* ‘squeeze’, *sak-in* ‘avoid/protect oneself from’).¹ If the set of Roots capable of forming TVRs is not predictable solely

on the basis of lexical semantics, then any analysis must enumerate this set of Roots explicitly; on our analysis, selectional restrictions will regulate the combination of reflexive Voice with particular Root+*v* combinations (see section 5).

As for thematic restrictions, viz. the fact that TVRs only derive agent-oriented reflexives, this type of restriction is robust in Turkish. Towards the end of the paper, we discuss how it follows from an account with Reflexive Voice, and how our specific account may generate predictions on the range of roles that can be Voice-identified with the agent role.

2 Intransitivity and monadicity

Numerous diagnostics distinguish transitive verbs with pronominal reflexive arguments from verbal reflexives; we provide a select few indicative tests here. Some are interpretive tests, teasing apart dyadic from monadic predicates (Abusch, 1989; Reuland, 2001; Reinhart and Siloni, 2004, 2005; Reinhart and Reuland, 1993; Jackendoff, 1992; Labelle, 2008; Safir, 2004; Lidz, 2001b; Marelj and Reuland, 2016; Dimitriadis and Everaert, 2014; Reuland and Winter, 2009; Sells et al., 1987; Reuland, 2018; Zec, 1985); other tests are sensitive to syntactic transitivity (here, the behavior of reflexives under causativization). In Turkish, the two classes of tests converge: transitive verbs with pronominal reflexive arguments consistently pattern as dyadic and transitive, and TVRs as monadic and intransitive.²

2.1 Interpretive diagnostics

VP ellipsis treats pronominal and verbal reflexives differently in Turkish: an ellipsis site anteceded by the verb *hazırla-* ‘prepare’ taking a pronominal reflexive object, (6a), licenses both strict and sloppy readings, suggesting that the reflexive pronoun is obligatorily recovered at the ellipsis site. Things work differently in the verbal reflexive counterpart (6b), where only the sloppy reading is available. If the verbal reflexive were secretly a transitive, this contrast should not arise. This differential availability of strict/sloppy ambiguities can also be seen in comparative ellipsis and in the different denials available to assertions modified by *only* (see Dahl 1973; Sauerland 2013; Sportiche 2014).

- (6) a. Ali kendin-i hazırla-dı, Ayşe de (öyle yap-tı).
 Ali self-ACC prepare-PST Ayşe too so do-PST
 ‘Ali prepared himself, and Ayşe did so too.’ ✓strict ✓sloppy
- b. Ali hazırla-n-dı, Ayşe de (öyle yap-tı).
 Ali prepare-REFL-PST Ayşe too so do-PST
 ‘Ali prepared, and Ayşe did so too.’ ✗strict ✓sloppy

Another diagnostic comes from a type of *deferred reference* (e.g. Fauconnier, 1985; Abusch, 1989; Jackendoff, 1992; Reinhart and Reuland, 1993). Under particular circumstances, instead of picking

out the antecedent itself, the reflexive element can instead pick out a contextually salient proxy for the antecedent. Such cases of relaxed identity – termed proxy readings – have been noted to differentiate between different reflexivity strategies cross-linguistically, variously distinguishing between verbal and pronominal reflexives and/or between different kinds of pronominal reflexives (see e.g. Reuland 2018 for a summary, and for analyses see Reuland and Winter 2009; Safir 2004; Raghotham 2024; Paparounas 2023; pronominal reflexives of the so-called SE type seem to not behave uniformly in this test, see Marelj and Reuland 2016 for recent discussion and references). In Turkish, proxy readings clearly distinguish between pronominal and verbal reflexives: the pronominal anaphor *kendi* in (7a) felicitously picks out the proxy in the given context, while the verbal reflexive in (7b) is infelicitous.

- (7) Context: Kivanç sees that his wax statue is getting very dusty, and decides to dust it off.
- | | | | |
|----|---|----|---|
| a. | Kİvanç kendin-i silkele-di.
Kİvanç self-ACC shake-PST
'Kİvanç shook himself off.' | b. | #Kİvanç silkele-n-di.
Kİvanç shake-REFL-PST
'Kİvanç shook (himself) off.' |
|----|---|----|---|

These facts furnish a further interpretive argument against assimilating TVRs to their pronominal counterparts. Turkish pronominal reflexives clearly involve a transitive syntax: the reflexive DP argument is syntactically distinct from its antecedent. If verbal reflexives similarly involved a transitive syntax with two distinct DPs, we would expect them, too, to give rise to human-acts-on-proxy interpretations, but this is not what we find.

2.2 Syntactic diagnostic: Indirect causatives

Both verbal and pronominal reflexives can be causativized with an *indirect causative* interpretation, which is allowed in Turkish only with predicates that have an external argument (Akkuş 2021, 2023; see also Zimmer 1976). Crucially, in Turkish the case of the causee depends on the transitivity of the causativized base. When a transitive predicate is causativized, the causee bears dative case, whereas in causativization of an unergative intransitive the causee carries accusative case (Lewis 1967; Kornfilt 1997; Göksel and Kerslake 2005; Çetinoğlu et al. 2008; Akkuş 2021, a.m.o).

Predicates with pronominal reflexive arguments pattern as regular transitives: the causee bears dative, (8)-(9). But when a verbal reflexive is causativized, the causee bears accusative case, (10) (cf. Kayne 1975, 203-204, 393 for French *se* reflexives).

- (8) Çocuk kendin-i besle-di. (9) Ebeveynler-i çocuğ- {*a* / **u*} kendin-i besle-t-ti.
 child self-ACC feed-PST parents-3POSS child {DAT / *ACC} self-ACC feed-CAUS-PST
 ‘The child fed himself.’ ‘His parents made the child feed himself.’

- (10) a. *pro* bu ara berbat besle-n-iyor-um.
 this while terrible feed-REFL-PROG-1SG
 ‘I feed (myself) terribly these days.’
- b. [*The speaker complains that his wife doesn’t cook, leading to lots of take-out food.*]
 Merve {ben-i / *ban-a} bu ara berbat besle-n-dir-iyor.
 Merve {I-ACC / *I-DAT} this while terrible feed-REFL-CAUS-PROG
 ‘Merve is making me feed (myself) terribly these days.’ <https://tinyurl.com/mrx2zdes>,
 12:40.

That TVRs always appear with an accusative causee is the behavior expected if they are intransitive.

Overall, TVRs behave as monadic predicates interpretively and as intransitives for the syntactic causativization diagnostic; they thus pattern distinctly from pronominal reflexives, which are interpretively dyadic and syntactically transitive. Recall now that the diagnostics we have just deployed are sensitive to valence, but indifferent as to the position of arguments. Turning to this remaining issue for TVRs, we find a striking instance of mixed behavior: the surface subject behaves as a deep object for some syntactic diagnostics, and as a deep subject for others.

3 Internal argumenthood

Various tests show that the surface subject of TVRs originates as an internal argument. These tests distinguish unaccusatives/passives from unergatives/transitives in the language. In this section, TVRs pattern consistently only with unaccusatives and passives, and not with unergatives; whenever TVRs pattern with transitives, their surface subject patterns with transitive objects, not transitive subjects.

For the sake of space, we omit certain convergent diagnostics: for instance, the behavior of restitutive readings of the adverb *tekrar* ‘again’ in TVRs suggests the existence of a constituent that includes the verb and the surface subject but excludes the locus of agent entailments, a situation only possible if the surface subject originates low.

3.1 Resultatives

Nominal and adjectival resultatives crosslinguistically serve as a strong indicator of deep objecthood (e.g., Simpson 1983; Levin and Rappaport Hovav 1995). In Turkish, resultatives display the requisite baseline behavior. Firstly, they can only be predicated of direct objects of transitives, as well as surface subjects of passives and unaccusatives (11); crucially, resultatives predicated of subjects of unergatives are systematically impossible, (12).

- (11) a. Leyla masa-yı (ter-temiz) sil-di.
 Leyla table-ACC REDUP-clean wipe-PST
 ‘Leyla wiped the table (completely clean).’ (Turgay 2013:(27a)) *transitive*

- b. Masa (Leyla tarafından) ter-temiz sil-in-di.
 table (Leyla by) REDUP-clean wipe-PASS-PST
 ‘The table was wiped completely clean (by Leyla).’ *passive*
- c. Nehir kas-katı don-du.
 river REDUP-solid freeze-PST
 ‘The river froze solid.’ (Turgay 2013:112,(51a)) *unaccusative*

- (12) *Ceren yop-yorgun koş-tu.
 Ceren REDUP-tired run-PST
 ‘Ceren ran tired.’ (based on Turgay 2013:69, (24b)) *unergative (predicate OK as depictive)*

Secondly, Turkish resultatives are not compatible with unselected objects, (13), or ‘fake reflexives’ of the English type (Turgay, 2013, 69).

- (13) *Leyla demliĝ-i boş iç-ti.
 Leyla teapot-ACC empty drink-PST
 Intended: ‘Leyla drank the teapot empty.’ (Turgay 2013:89, (27b))

They also cannot be predicated of unprojected elements. In out-of-the-blue contexts, resultatives predicated of missing objects are not licit, (14); they are only licit in *pro*-dropped contexts, (15).

- (14) a. Q: Ne ol-du?
 what happen-PST
 ‘What happened?’
 b. A: Leyla *(masa-yı) ter-temiz sil-di.
 Leyla table-ACC REDUP-clean wipe-PST
 ‘Leyla wiped *(the table) completely clean.’
- (15) a. Q: Leyla masa-ya ne yap-tı?
 Leyla table-DAT what do-PST
 ‘What did Leyla do to the table?’
 b. A: Sadece *pro*_{obj} ter-temiz sil-di.
 only REDUP-clean wipe-PST
 ‘She just wiped it completely clean.’

Overall, the baseline distribution of Turkish resultatives tracks the presence of projected deep objects.

Crucially, resultatives can be freely predicated of the surface subject of verbal reflexives; see (16), with its pronominal reflexive counterpart given in (17). Another example is provided in (18).

- (16) Ter-temiz yıka-n-dı-m.
 REDUP-clean wash-REFL-PST-1SG
 ‘I washed myself clean.’
- (17) Kendi-m-i ter-temiz yıka-dı-m.
 self-1SG.POSS-ACC REDUP-clean wash-PST-1SG
 ‘I washed myself clean.’
- (18) “...Bir sürü Himba insanı görebiliyorsunuz, özellikleri ... (you can see a number of Himba folks, their distinctive feature is ...)”
 ... kıp-kırmızı-ya boya-n-mış ol-ma-lar-ı.
 ... REDUP-red-DAT paint-REFL-PTCP be-INF-PL-POSS
 ‘them having painted themselves completely red.’ <https://tinyurl.com/5yea9vcn>, 15:45.

TVRs thus have an internal argument position, occupied underlyingly by the surface subject.³

3.2 Adjectival participles

A further diagnostic suggesting a low origin for the surface subject of TVRs comes from two types of stative (‘adjectival’) participles in Turkish. Studies of stative participles propose two kinds of analyses of their argument structure. One kind of analysis posits a low origin for the DPs appearing with stative participles, such that a participle is (minimally) a stativized *vP* that includes an internal argument (e.g. [Anagnostopoulou, 2003a](#); [Embick, 2004a](#); [Bešlin, 2022](#)). A different type of analysis posits that stative participles have their arguments introduced externally to the *vP*, on a par with adjectival predication (e.g. [Embick 2023](#), [McIntyre 2013](#), [Paparounas 2023](#), ch. 4). It is crucial in what follows that the participles of Turkish can be independently shown to instantiate the former structure, with a *vP* that includes an internal argument.

One diagnostic showing that Turkish stative passives embed ‘transitive’ (object-ful) *vPs* comes from passivizable verb-object idioms. Begin with (19), which shows that an idiom like ‘to take the bean out of the mouth’ meaning ‘to disclose a secret’ survives in eventive passives. On standard assumptions about idiom locality, this state of affairs is understandable if the idiomatic reading arises freely with any *vP* that includes the idiom’s fixed parts; as such, the availability of the idiom can be used to diagnose whether the DP *the beans* originated in the same *vP* as the verb *take* in a given case.

- (19) a. Sonunda bakla-yı ağız-dan çıkar-dı.
 finally bean-ACC mouth-ABL take.out-PST
 ‘S/he finally disclosed the secret.’
 b. B20 toplantı-sın-da bakla ağız-dan çıkar-ıl-dı
 B20 meeting-CM-LOC bean mouth-ABL take.out-PASS-PST
 ‘The secret was disclosed at the B20 meeting.’ ([Akkuş 2021:253,\(533\)](#))

Against this background, note that adjectival participles in *-mİş* license the idiomatic reading just like eventive passives do; (20). The participle thus must have its DP argument originating as a deep object.

- (20) Ağız-dan çıkar-ıl-mış bakla-lar huzur-u boz-du.
 mouth-ABL take.out-PASS-PTCP bean-ACC peace-ACC spoil-PST
 ‘The disclosed secrets have upset the peace.’

We accordingly follow much previous work in employing the *-mİş* participle as a low origin diagnostic for Turkish ([Nakipoğlu-Demiralp, 1998, 2002](#); [Acartürk, 2005](#); [Gürer, 2014](#)) and languages like it in the relevant respects ([Bhatt and Embick, 2004/2017](#); [Alexiadou and Schäfer, 2013](#)).

The pairs in (21) show that the *-mİş* participle is compatible with Roots that typically form unaccusatives (21a), but not unergatives (21b).

- (21) a. {bozul-muş / çürü-müş / eri-miş / kok-muş} yiyecek
decomposed-PTCP / rot-PTCP / melt-PTCP / smell-PTCP food
'the {decomposed/rotten/melted/smelling} food'
- b. {*koş-muş / *yüz-müş / *çalış-mış / *bağır-mış} çocuk
run-PTCP / swim-PTCP / work-PTCP / shout-PTCP child
Intended: 'a {run/swum/worked/shouted} child' (Nakipoğlu-Demiralp 2002:7a-b)

Moreover, typical transitive-forming Roots allow the formation of participles in *-miş*:

- (22) a. özensizce kes-il-miş kağıt-lar b. giy-il-miş pantolon
sloppily cut-PASS-PTCP paper-PL dress-PASS-PTCP pants
'sloppily cut papers' (Gürer 2014:29a) 'dressed/put on pants.'

Importantly, TVRs are perfectly compatible with *-miş*. In (23), we show that it is possible to produce a stative that appears outside the nonactive/reflexive morpheme, and produces a stative-of-a-reflexive-event interpretation. Since we know independently that the *v*Ps that form the input to stativization are those *v*Ps that involve internal arguments, the grammaticality of these examples speaks in favor of a low origin analysis for those *v*Ps that come to be reflexivized before they are stativized.

- (23) a. ört-ün-müş kadın b. eğ-il-miş çocuk
cover-REFL-PTCP woman bend-REFL-PTCP child
'a self-covered woman' 'the self-bowed child' (Gündoğdu, 2017, 99)

The same conclusion arises from the language's other participial formation, formed with the stativizer *-Ik* (Nakipoğlu-Demiralp 1998; Acartürk 2005; Acartürk and Zeyrek 2010; Neu 2026). In (24a) the *-Ik* participle combines with unaccusative-forming Roots, while it fails to do so in (24b) with unergative-forming ones.

- (24) a. kır-ık bardak b. *uyu-k bebek
break-ADJ glass sleep-ADJ baby
'the broken glass' Intended: 'the slept baby'

The *-Ik* participle is also compatible with anticausative idioms whose sole argument originates in a low position, (25) (the idiom's transitive counterpart can be passivized). Importantly, TVRs can again be the input to the formation of participles (26).

- (25) a. Adam-ın öd-ü kop-tu.
man-GEN bile-POSS break-PST
'The man got very scared.' (Lit: 'the man's bile broke'; Göksel and Öztürk 2019, 165)

- b. Adam-ın öd-ü kop-uk i-di.
man-GEN bile-POSS break-ADJ COP-PST
'The man was very scared.' (Lit: 'the man's bile was in a broken state.')
- (26) a. Herkes bedence cıvıl, ruhça ört-ün-ük.
everybody bodily naked soul-wise COVER-REFL-ADJ
'Everybody is bodily naked but soul-wise self-covered.' (Kaynak 2012:75)
- b. Korku-lar-ımız-a sığ-ın-ık yaş-ıyor-uz.
fear-PL-1PL.POSS-DAT squeeze-REFL-ADJ live-PROG-1PL
'We live self-refuged in our fears.' (Twitter, @hulyakrt, 9/23/21)

4 External argumenthood

The diagnostics in the previous section show that the surface subject in TVRs originates low, on a par with the surface subject of passives and unaccusatives. We now demonstrate that the sole argument also passes external argumenthood diagnostics, of the kind that cannot be attributed only to the agentive interpretation of reflexives or to grammatical subjecthood. Putting the diagnostics together reveals that a single argument is associated with two distinct positions inside VoiceP.

Some of the tests examined in this section are traditionally used to contrast transitives and unergatives from unaccusatives in Turkish. Applying these tests to TVRs, we find that their behavior breaks with the patterns established in section 3: here, TVRs pattern with unergatives as opposed to passives/unaccusatives, and their surface subject patterns with the subject, not the object, of transitives.⁴

Crucially, TVRs also pattern distinctly from passives for all relevant tests below. This divergence is important: it clarifies that the fact that reflexives sometimes pattern as external-argument-bearing verbs, and distinctly from unaccusatives, is not attributable solely to the presence of agentive semantics, which reflexives share with passives but not with unaccusatives (cf. Key 2023). Instead, the initially low surface subject of TVRs must transit through Spec,VoiceP, thereby sharing traits with 'real' external arguments for the purposes of the relevant tests. Importantly, we show in section 4.5 that the surface subject of TVRs passes internal and external argumenthood diagnostics in the same example, clarifying that we are indeed dealing with movement, and not structural ambiguity.

4.1 Episodic impersonals

For a majority of Turkish speakers, the addition of the 'passive' exponent to any predicate besides transitives with a thematic subject and a structurally case-marked object leads to an *impersonal* construction (Akkuş 2021; Legate et al. 2020). Unergatives and transitives can form impersonals in both generic and episodic contexts (27)-(28) (Nakipoğlu-Demiralp 2001; Acartürk 2005; Acartürk and Zeyrek 2010; Akkuş 2021; Legate et al. 2020).

- | | |
|--|---|
| <p>(27) <i>Unergative</i></p> <p>a. Her gece dans ed-il-ir.
every night dance do-IMPERS-AOR
'People/one dance(s) every night.'</p> <p>b. Konser boyunca uyu-n-du.
concert during sleep-IMPERS-PST
'People/one slept during the concert.'</p> | <p>(28) <i>Transitive with oblique object</i></p> <p>a. Otobüs-e bin-il-ir.
bus-DAT board-IMPERS-AOR
'People/one board the bus.'</p> <p>b. Otobüs-e bin-il-di.
bus-DAT board-IMPERS-PST
'People/one boarded the bus.'</p> |
|--|---|

On the other hand, impersonals of unaccusatives, while felicitous under a habitual reading, (29a), cannot receive an episodic interpretation (29b).

- (29) a. Türkiye-de her gün trafik kaza-lar-ın-da öl-**ün**-ür.
Turkey-LOC every day traffic accident-PL-CM-LOC die-**IMPERS**-AOR
'In Turkey people die in traffic accidents every day.' (Nakipoğlu-Demiralp 2001:140,(22b))
- b. *Dün burada öl-**ün**-dü.
yesterday here die-**IMPERS**-PST
Intended: 'People/one died here yesterday.' (Unaccusative)

As noted in Akkuş 2021 and Legate et al. 2020, impersonals involve an unpronounced impersonal pronoun (IMP) filling an argument position – they thus lack argument 'demotion'. The position occupied by the impersonal pronoun can be that of the thematic subject, as in (27)-(28), or the thematic object (29). Crucially, the availability of an episodic interpretation is modulated by the syntactic position of IMP: episodic interpretations are allowed with IMPs in Spec,VoiceP, but disallowed with IMPs merged as internal arguments.⁵

TVRs pattern with unergatives and transitives in being compatible with both habitual and episodic aspects. Consider the examples in (30) with episodic aspect:

- (30) a. Bu çim-ler-e uza-n-ıl-mış.
this grass-PL-DAT lie.down-REFL-**IMPERS**-PST
'Apparently people/one laid (themselves) down on the grass.'
- b. Büyük-ler-in ön-ün-de saygı-yla eğ-il-in-di.
elder-PL-GEN front-POSS-LOC respect-with bend-REFL-**IMPERS**-PST
'People/one bowed respectfully in front of the elder people.'

To repeat, verbal reflexives pattern with 'active' structures for the purposes of impersonals, in which a syntactically projected argument is found in Spec,VoiceP.

4.2 Agent nominalization

The availability of agent/external argument nominalization has sometimes been used as a diagnostic of the argument structure of verbal reflexives; see e.g. Reinhart and Siloni (2005); Alexiadou and Schäfer (2013); Amato (2024); Paparounas (2023). Turkish provides the appropriate baseline to apply

the test: agent nominalization is sensitive to the unergative/unaccusative distinction, independently of verbal reflexives. Thus, Roots that typically form unergatives/transitives freely form agent nominals in *-IcI*, (31a). Unaccusative-forming Roots, however, resist agent nominalization, (31b) (see Nakipoğlu-Demiralp 1998; Acartürk 2005). (31c) is a particularly probative example: it shows that a typical unaccusative-forming Root, in this case $\sqrt{\text{FINISH}}$, can only yield an agent nominal if it is first transitivized.

- | | | | |
|------|------------------------------------|---|--|
| (31) | a. koş-ucu
run-NMLZ
'runner' | b. *düş-ücü
fall-NMLZ
Int: 'faller' | c. bit-*(ir)-ici
finish-CAUS-NMLZ
'finisher' |
|------|------------------------------------|---|--|

Consider now the attested examples in (32) showing that TVRs can be the input to agent nominalization; crucially, these are *not* Root nominals, witness the surfacing of Voice morphology between the Root and the nominalizer:

- (32) a. "... eğer padişahlığ-a er-ecek ol-ur-sa-n haram-dan sak-ın-ıci ol."
if sultanate-DAT reach-FUT be-AOR-COND-2SG sin-ABL save-REFL-NMLZ be
'If you reach the sultanate, be an avoider of sin [lit. a self-protector-from sinning].' (Şahin 2019:79)
- b. Putin Hindistan-da. Seri sar-ıl-ıci Modi tabii on-u da kucakla-dı.
Putin India-LOC serial wrap-REFL-NMLZ Modi of.course he-ACC also embrace-PST
'Putin is in India. The serial hugger (lit: self-wrapper) Modi of course gave him a hug too.' (Twitter, @OmerFarukGorcin, 10/5/18)

Previous literature has sometimes claimed that agent nominalizations of TVRs are not well-formed (e.g., Gündoğdu 2017). As far as we can tell, such claims result from individual examples having been judged in isolation and without context. We have provided attested examples here, additionally judged as grammatical by our consultants, to help control for the kinds of pragmatic factors that are known to affect the acceptability of derived nouns more generally. Firstly, the agent nominal must be afforded a specialized usage in lexicosemantic space; this is often achieved via modification (e.g. *serial hugger* in (32b); cf. e.g. Embick and Marantz 2008 on English *#stealer* versus *base-stealer*). Secondly, agent nominals must denote entities that are *name-worthy* in a sense familiar from the literature on noun incorporation (see e.g. Carlson, 2006, 10ff) and implicit indefinites (see e.g. Martí, 2015), describing activities institutionalized enough to be worthy of a label (e.g. *dile-n-[i]ci* 'beggar (lit: self-wisher)' is a well-established agent nominal constructed from a verbal reflexive). TVRs then can, in fact, undergo agent nominalization in the general case, though judgments on individual formations are often modulated by grammar-external conditions of appropriateness of use.

The capacity to form agent nominals is thus another trait that TVRs share with unergatives, deviating from passives and their ilk. Various examples with agent nominals further highlight the dissociation between passives and reflexives. For example, the verb *sar-ıl* ‘wrap-NACT’ in (32b) is ambiguous between passive (‘to be wrapped around’) and reflexive (‘to wrap oneself around’) interpretations in finite contexts. But the addition of the agent nominalizer *-IcI* leaves behind only the reflexive interpretation: the form *sar-ıl-ıcı* cannot mean ‘one who gets hugged/wrapped around’, but only ‘one who hugs/wraps oneself around (someone)’.

4.3 Adverbial gerundive *-ArAk*

In this section we consider adverbial gerundives with the verbal suffix *-ArAk* (Özkaragöz 1980, Knecht 1985, Biktimir 1986, Kornfilt 1997, Legate et al. 2020, Akkuş 2021, Akkuş and Paparounas 2024). *-ArAk* clauses display intricate matching restrictions that place verbal reflexives in the same natural class as active transitives/unergatives, distinct from passives or unaccusatives. Briefly, *-ArAk* gerunds necessitate that the matrix and gerundive clause match in voice, in the status of the subject as underlying or derived, or a combination of both. Table 1 summarizes the relevant baseline patterns.⁶

Transitive/unergative + transitive/unergative	✓
Unaccusative + unaccusative	✓
Unergative + unaccusative	✗
Passive + passive	✓
Passive + transitive/unergative	✗
Passive + unaccusative	✗
Verbal reflexive + transitive/unergative	✓
Verbal reflexive + unaccusative	✗
Verbal reflexive + passive	✗

Table 1: *Patterns of combinations with -ArAk*

Verbal reflexives behave like unergatives/transitives, (33a-33b), in being able to combine with each of these classes but not with unaccusatives, (33c).

- (33) a. Kız [söyle-n-erek] yürü-dü.
 girl say-REFL-ARAK walk-PST
 ‘The girl walked (while) complaining.’ (reflexive + unergative; Nakipoğlu-Demiralp 2002, 13c)
- b. Adam [söyle-n-erek] bulaşık-lar-ı yıka-dı.
 man say-REFL-ARAK dish-PL-ACC wash-PST
 ‘The man did the dishes (while) complaining.’ (reflexive + transitive)

- c. *Adam [buna-yarak] yıka-n-dı.
 man go.senile-ARAK wash-REFL-PST
 ‘The man washed (while) going senile.’
 The man was washed (while) going senile. (unaccusative + reflexive)

Moreover, reflexives are not compatible with passives, clarifying that the mere presence of an agentive entailment in reflexives does not suffice to explain why reflexives do not combine with unaccusatives.

- (34) a. *Manken [giy-in-erek]öp-ül-dü.
 model dress.up-REFL-ARAK kiss-PASS-PST
 ‘The model was kissed (while s/he was) getting dressed up.’
 b. *Çocuk [okşa-n-arak] söyle-n-di.
 child caress-PASS-ARAK say-REFL-PST
 ‘The child complained (while s/he was) being caressed.’

In terms of *-ArAk* clauses then, reflexives pattern with unergatives and transitives (which have a filled Spec,VoiceP), and distinctly from both unaccusatives (which lack a thematic Voice layer altogether), and from passives (which have a thematic Voice, albeit one lacking a specifier).

4.4 Non-passivization

Verbal reflexives may not be passivized in Turkish, and instead result in impersonals (Legate et al. 2020; Akkuş 2021) - a fact again expected if they behave, for the purposes of this diagnostic, like active Voice with an argument occupying Spec,VoiceP, and not passive Voice (cp. e.g. Schäfer, 2012).

Sequences of Refl+‘Pass’ (observed in the prior literature, e.g., Göksel 1993) are ungrammatical when a *by*-phrase is added, (35). The addition of the ‘passive’ morpheme leads to an *impersonal* interpretation, and we accordingly translate the relevant examples as ‘People/one self-Verbed’.

- (35) a. Dün dere-de (*adamlar tarafından) yıka-n-ıl-dı.
 yesterday river-LOC men by wash-REFL-IMPERS-PST
 ‘People/one self-washed in the river yesterday (*by some men).’
 b. Sıcak ol-duğ-u için (*tatilciler tarafından) soy-un-ul-du.
 hot be-NMLZ-POSS for vacationers by undress-REFL-IMPERS-PST
 ‘People/one self-undressed (*by the vacationers) since it was hot.’

Further confirmation of the impersonal reading comes from the fact that TVRs with the ‘passive’ morpheme require a human interpretation, (36), and that only 3sg agreement is allowed, (37).

- (36) a. Bu orman-da {çocuk-lar / fil-ler} dere-de yıka-n-ır-lar.
 this forest-LOC {child-PL / elephant-PL} river-LOC wash-REFL-AOR-PL
 ‘In this forest, {children / elephants} wash (themselves) in the river.’

- b. Bu orman-da dere-de yıka-n-ıl-ır.
 this forest-LOC river-LOC wash-REFL-IMPERS-AOR
 ‘In this forest, {people / #elephants} wash (themselves) in the river.’
- (37) *Hamam-da yıka-n-ıl-dı-m.
 Turkish bath-LOC wash-REFL-PASS-PST-1SG
 Intended: ‘Self-washing was done by me in the Turkish bath.’

The non-passivizability of verbal reflexives is instructive in indicating the presence of an argument in Spec,VoiceP. The crucial observation is that verbal reflexives behave in the same way as impersonals. Following the analysis of impersonals in [Akkuş 2021](#) and [Legate et al. 2020](#) (cf. section 4.1 for a summary), the absence of passivization straightforwardly follows if the IMP occupies the specifier of VoiceP, such that TVRs pattern like active transitives or unergatives for this test.

The patterns in this section also highlight that the behavior of TVRs in causatives, noted in section 2.2, provides an additional external argumenthood test. Since indirect causatives embedding verbal reflexives are only possible with structures that have a filled Spec,VoiceP in Turkish (see [Akkuş and Paparounas 2024](#)), the fact that verbal reflexives form indirect causatives yields another argument that their initially low surface subject transits through Spec,VoiceP.

4.5 External-Internal properties combined

The surface subject of TVRs can simultaneously pass internal and external argument diagnostics in the same example. In (38a), the reading of an arbitrary human impersonal in the episodic context is reserved for external arguments; but the resultative points to the presence of an internal argument. Likewise, in (38b), causativization signals the presence of an argument in embedded Spec,VoiceP (see sections 2.2 and 4.4), with the resultative again suggesting the presence of a deep object.

- (38) a. Dün bu dere-de ter-temiz yıka-n-ıl-dı.
 yesterday this river-LOC REDUP-clean wash-REFL-IMPERS-PST
 ‘People/one washed clean yesterday.’
- b. *pro* [çocuğ-u ter-temiz yıka-n]-dır-dı-m.
 [child-ACC REDUP-clean wash-REFL]-CAUS-PST-1SG
 ‘I made [the child wash clean].’

Such combinations call for a structure that can accommodate the internal and external argument properties of the same argument in the same example; they thus rule out the possibility that TVRs are ambiguous between unergative and unaccusative structures. Lability of this kind of course arises quite generally elsewhere (for unergative/unaccusative alternations, see e.g. [Perlmutter 1978](#); [Burzio 1981, 1986](#); [Sorace 2000](#); [Krejci 2020](#); [Holisky 1987](#); [Neu 2026](#)) and arguably with reflexives in other languages (see esp. [McGinnis 2022](#) for *se* reflexives); but it does not suffice to explain our data. Besides

suggesting the necessity of a movement analysis as opposed to mere structural ambiguity, examples like (38) reinforce a point already evident in our discussion up to this point: different phenomena can be sensitive to distinct positions forming part of the same movement chain. In (38), it suffices for the licensing of the resultative that the surface subject originates as a deep object. For the purposes of impersonals and causatives, by contrast, it suffices that the same argument ended up occupying Spec,VoiceP on its way to becoming the grammatical subject. This type of differential sensitivity of particular phenomena to particular positions along a movement chain is familiar from other domains (see e.g. McCloskey 1997 on the base position of ‘subjects’; Levin and Rappaport Hovav 1995, 17ff for the surface subjects of unaccusatives; and Kastner 2017, 17ff for relevant discussion with specific reference to reflexives).⁷

5 Analysis: Reflexive Voice and Movement

5.1 Syntax

The Turkish facts we have surveyed up to this point call for a mechanism of reflexivization capable of deriving a reflexive interpretation from a single-argument structure, one where the single argument in question has a basic position VP-internally and comes to occupy a derived position VoiceP-internally before attaining its ultimate position as grammatical subject.

Section 2 showed that TVRs are not transitive verbs. We must thus derive a reflexive interpretation from an intransitive structure, without resorting to a (null) reflexive pronoun in the syntactic structure of these verbs. In unergative analyses of reflexives – where the sole argument is base-generated externally to the *v*P (see e.g., Chierchia 2004; Bruening 2006; Reinhart and Siloni 2004, 2005; Labelle 2008; Jo 2019) – this is an easy task: with an appropriately typed *v*P, a Voice head could effectively ‘pull out’ the unsaturated theme role and identify it with a distinct element. But deriving a reflexive denotation from a basic unaccusative structure is an underappreciated challenge, since identifying the entity already mapped to theme with a distinct element in a non-stipulative way is non-trivial; we refer the reader to Paparounas (2023, 115ff) for a more detailed exposition of the issue. Existing unaccusative analyses, when explicit on the semantics, resort to exceptional placements of lambda abstracts (see e.g. Spathas et al. 2015, cf. Barker 2007), departures from standard assumptions on the interpretation of movement (e.g. Kastner 2017), or direct identification of thematic roles (Paparounas, 2023; Oikonomou and Alexiadou, 2022), a move unclear in its formal details.

We note also that in Turkish, as in other languages with similar voice systems, verbal reflexivization seems strongly tied to grammatical voice: TVRs surface with a particular voice morphology, and that morphologically (and syntactically) active verbs never yield verbal reflexives, as noted in §1.

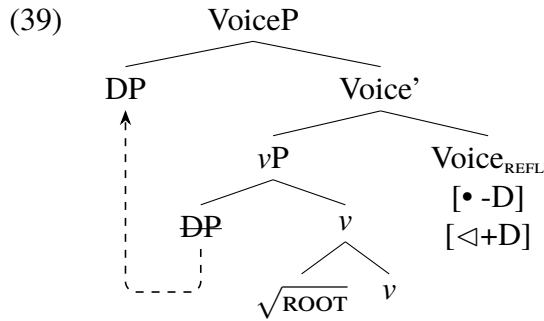
We accordingly propose to link reflexivity to the encoding of grammatical voice, which we take to be carried out in large part by the head Voice, which determines the properties of the external argument. We posit a reflexive variant of Voice (following Labelle 2008; McGinnis 2022; Raghotham 2022; Oikonomou and Alexiadou 2022; Paparounas 2023; Kallulli and Roberts 2025; cf. Spathas et al. 2015; Alexiadou 2014; and for partially related ideas in anaphor binding, see e.g. Ahn 2015; Kratzer 2009; Paparounas and Akkuş 2024). The move in these works is to treat ‘verbal’ reflexivity as a kind of external argument introduction; it is this type of analysis we propose for Turkish. Effectively, if active Voice introduces an agent function as well as a DP that saturates it, and passive Voice an existentially closed agent function, reflexivity can also be understood as a kind of agent introduction, one where the agent role ultimately comes to be identified with a distinct role.

Recall from section 1 that the set of Roots capable of forming TVRs is neither small nor lexico-semantically circumscribed. In our analysis, selectional restrictions must guarantee that the reflexivizing Voice head will appear only with those Roots, or Root+*v* combinations, capable of forming verbal reflexives in Turkish; in this it resembles other Voice heads (see section 1 as well as Alexiadou et al. 2006; Schäfer 2008; Alexiadou and Schäfer 2013). There are different ways of implementing this selectional relationship: for instance, it could be that the identity of the Root is visible for selection at the *v*P level (see Merchant 2019 for categorizers and Roots operating jointly in selection), or the relevant Roots could bear a diacritic feature visible at the *v*P level. Since the choice between such alternatives would leave our analysis otherwise unaffected, we note only that the existence of such restrictions (however implemented) is unremarkable in light of other domains of argument structure. It is not uncommon cross-linguistically for there to be lexical restrictions on the formation of causatives (e.g. Pylkkänen, 2008, 107ff); anticausatives (e.g. Alexiadou et al., 2006; Schäfer, 2008); or passives (e.g. Doron, 2003; Alexiadou and Doron, 2012), with the set of Roots that do or do not appear in these structures typically not characterized solely on a lexicosemantic basis. Our contention here is simply that TVRs are no different from these other voice alternations.⁸

The basic syntax of Voice_{REFL} is characterized by its inability to introduce a DP. Following the notation in Müller (2010) and subsequent work, we notate this restriction as a feature [$\bullet$ -D] which prohibits external Merge of Voice_{REFL} with elements of category D. Voice_{REFL} shares this feature with other nonactive Voice heads in the sense of Alexiadou et al. 2015 (see also Schäfer 2008; Legate 2014; Legate et al. 2020): the Voice heads building passives and (marked) unaccusatives also bear [$\bullet$ -D], and it is this natural class that leads to the emergence of non-active morphology across these categories in Turkish (see Key 2024 for Turkish, and for broader discussion see Alexiadou et al. 2015;

Paparounas 2023; Embick 1997, 2004b; Kastner 2016; Oikonomou and Alexiadou 2022).

Though $\text{Voice}_{\text{REFL}}$ resembles other non-active Voice heads in its inability to be Externally Merged with a DP, it differs in demanding to be combined with a DP via Internal Merge. $\text{Voice}_{\text{REFL}}$ bears the trigger of the movement step diagnosed in sections 3-4: we take this to be an Internal-Merge-triggering feature [$\triangleleft$ D] (following Hewett 2023), as in (39) below. This is an EPP feature posited independently in work on different phenomena (see e.g. Hopperdietzel 2024 for Mandarin, Deal 2019 for Nez Perce, Lee 2024 for Korean, Sundaesan 2012 for Tamil; cf. Ahn 2015 for English anaphors; and see discussion of edge features elsewhere including Chomsky 2008; Abels 2012; Georgi 2017).

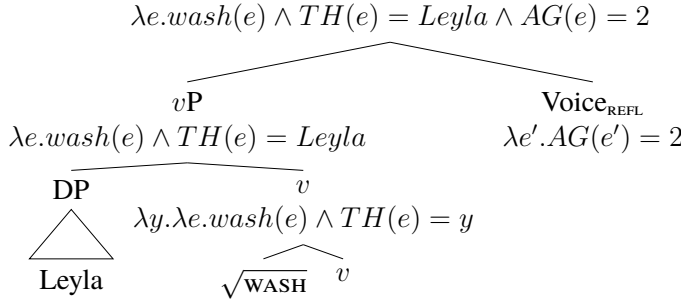


5.2 Interpretation

With this basic syntax in place, we propose an account of how reflexivization comes about on Voice; in doing so, we develop ideas on the precise interpretation of Reflexive Voice originally suggested to us by Kyle Johnson (p.c.), to whom we are indebted. We take thematic roles to be functions from events to individuals in a neo-Davidsonian event semantics (see e.g. Parsons, 1990), introduced by particular functional heads at LF, as is common in recent argument structure literature (e.g. Myler 2016; Wood 2015; Bruening 2013; Legate 2014; Alexiadou et al. 2015; Schäfer 2008; Borer 2005; for antecedents see Chomsky 1993, Baker 1997, 121-122, Reinhart 2016, 3). The theory we assume thus has no thematic positions in the strict sense, contrasting with a more traditional conception of θ -roles as entities in the syntax (or the lexicon-syntax mapping, Chomsky 1981).

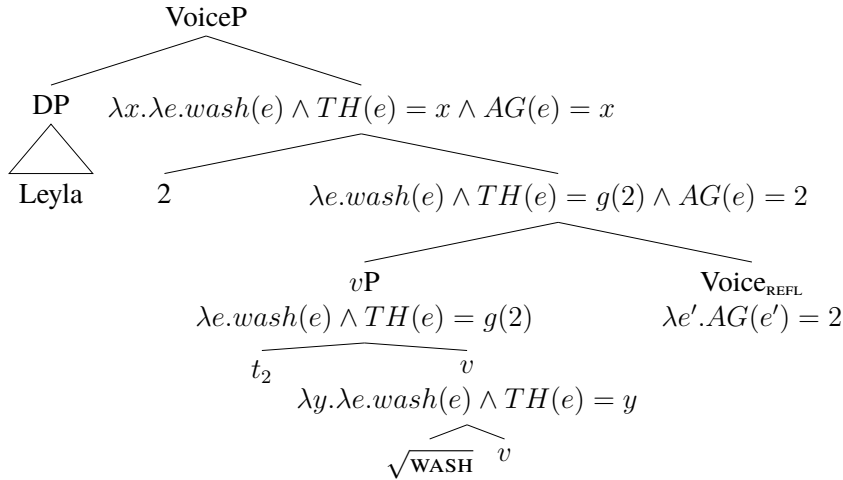
In (40), the denotation of $\text{Voice}_{\text{REFL}}$ introduces an Agent function, like other (thematic) Voice heads; but it also introduces a free variable, represented as a numerical index feature 2 (Heim and Kratzer 1998, Kratzer 2009). This free variable saturates the Agent function, providing an $\langle s, t \rangle$ denotation for the Voice head that can combine with the vP meaning via Predicate Modification.

(40)



For the denotation in (40) to ultimately be well-formed, the index 2 will have to be assigned an interpretation. We propose that this index can only be interpreted under semantic binding, made possible by movement of the internal argument. Our analysis is inspired by the Binder Index Evaluation Rule of [Büring \(2005\)](#), where quantified NPs come with an index demanding that pronouns c-commanded by such NPs and bearing the same index are necessarily bound pronouns. Our approach similarly suggests that the index 2 on $Voice_{REFL}$ is to be interpreted as a bound variable (see [Altshuler et al. 2019](#) for a concrete separation of indices into discourse-related and trace-related; also see [Kratzer 2009](#)). Once *Leyla* moves, it binds both its trace and the free variable introduced on $Voice$.

(41)



In (41), we follow the convention in [Heim and Kratzer \(1998\)](#), where the landing site of movement involves an index that triggers lambda abstraction. The lambda abstract triggered by the index 2 comes to bind both the trace bearing that index and the free variable on $Voice$.⁹

Note that representations like (41) are LF trees: we are assuming that they result from a set of principles mapping syntactic movement chains to configurations of semantic binding (see e.g. [Büring 2005](#), and for brief discussion linked directly to verbal reflexives, see [Reuland 2018](#), 90-91). In particular, we are assuming that LF translates lower copies of movement into variables (or larger structures embedding variables; see e.g. [Fox 1999](#); [Sauerland 2004](#); [Poole 2024](#)), and that lambda-abstraction-

triggering indices are effectively sprouted at the landing site of movement as just mentioned, yielding configurations like (41). In other words, though elsewhere adhering to the program in Heim and Kratzer (1998), we treat a substantial part of the machinery developed therein as obtaining at LF; in particular, we do not take indices to be present in the syntax proper (Chomsky, 1995; Reuland, 2011).¹⁰

5.3 *Consequences and comparisons*

Our approach successfully captures a range of important properties of TVRs.

Firstly, it successfully derives a reflexive interpretation from an unaccusative/passive-like structure; a non-trivial task, as outlined above. Secondly, the approach successfully captures the interpretive properties of TVRs, diagnosed in §2.1. The structure in (41) includes only a single syntactic argument, the DP *Leyla*; the index 2, like the trace of *Leyla*, is non-distinct from *Leyla* for the purposes of the interpretive phenomena. In particular, (41) does not employ an anaphoric argument to saturate a thematic role, as the derivation of a transitive verb with a pronominal reflexive would; rather, the inclusion of the free variable derives the effect of reflexivizing the agent role, once semantic binding of this variable takes place. The assumption that the free variable can be interpreted only under binding, and not coreference, derives the interpretive asymmetries between pronominal and verbal reflexives. Whereas Turkish pronominal reflexives can be interpreted either via coreference or via binding under ellipsis, the free variable on Voice is only interpretable under binding, and thus fails to allow both strict and sloppy readings; pronominal reflexive arguments are sufficiently distinct from their antecedents to license proxy readings, whereas the free variable is not (see Paparounas 2023; Raghotham 2022; Reuland and Winter 2009, ch. 3 for relevant discussion); and other interpretive limitations characteristic of TVRs but not shown here for reasons of space also follow (e.g. obligatoriness of *de dicto* readings with attitude verbs). Overall, the configuration we postulate builds a unary predicate out of the verb phrase by means of how the index on reflexive Voice is interpreted (cp. the notion of syntactic bundling in Reinhart and Siloni 2005); in all relevant respects, the free variable resembles the interpretation of lower copies of movement chains, which themselves never license strict/sloppy ambiguities or partial identity readings.

Thirdly, the association of reflexivity with Voice straightforwardly captures the fact that reflexivization in Turkish goes hand in hand with a particular kind of voice morphology. The fact that reflexives are voice-selective, appearing only with nonactive morphology, would be mysterious if reflexivization were orthogonal to Voice: concretely, we would have no explanation for why a morphologically active verbal form cannot be interpreted reflexively.¹¹

Further contrasts between pronominal reflexives and verbal reflexives follow. For instance, the Turkish pronominal reflexive can be bound long-distance, as in (42) (Kornfilt, 1997, 2001), but TVRs disallow this possibility, (43). Since theta-role identification takes place locally on the reflexive Voice head, the impossibility of a long-distance interpretation of TVRs naturally follows on our analysis.

- (42) Çocuk [anne-si kendi-ni yıka-yacak diye] duy-du.
 boy mother-his self-ACC wash-FUT that hear-PST
 ‘The boy_i heard that his mother_k will wash him_i/herself_k.’
- (43) Çocuk [anne-si yıka-n-acak diye] duy-du.
 boy mother-his wash-REFL-FUT that hear-PST
 ‘The boy_i heard that his mother_k will wash *him_i/herself_k.’

We conclude this section by comparing our analysis of TVRs and existing analyses of different kinds of reflexive verbs that capitalize on movement in different ways. The shape of the semantic derivation we propose is clearly reminiscent of the standard analysis of Romance clitic reflexives developed in e.g., Marantz 1984; Kayne 1988; Pesetsky 1995; McGinnis 2004: in this type of analysis, movement of a deep object DP to the grammatical subject position across a pronominal clitic argument introduced in the specifier of Voice ensures Condition A-compliant syntactic binding of the clitic anaphor and produces thematic role identification. Ultimately, the analyses differ. In Turkish, the crucial movement can be shown in Turkish to be VoiceP-internal, distinct from the ordinary promotion to grammatical subject undertaken by the deep object of passives and unaccusatives without the intermediate movement step that characterizes TVRs. Moreover, since there is no pronominal reflexive argument in Turkish (see section 2.1, our analysis posits no syntactic binding, but only semantic binding of the free variable contributed by Voice.

Prima facie, our analysis is closer in spirit to the proposals for movement-driven reflexivization in Hornstein (2001); Boeckx et al. (2008). The analysis of English *shave*-type verbs in Hornstein (2001, 41), schematized in (44), bears an obvious initial resemblance to our analysis of Turkish.

- (44) [_{TP} John [_{VOICEP} John [_{VP} shaved John]]].


In substance, the specifics of our analysis ultimately diverge from these existing proposals in important ways. The theory of construal in Hornstein (2001), further specified for different kinds of reflexive dependencies in Boeckx et al. (2008), involves saturation of two distinct roles by one and the same argument (cf. Grohmann 2003, ch. 3). Under the traditional view of thematic roles presupposed in these works, this necessitates abandoning (or relaxing) the *Theta Criterion* (Chomsky 1981; see Collins 2024 for a contemporary view). In our analysis, there is no technical sense in which

the same entity saturates two separate thematic functions. The displaced internal argument ends up semantically binding the free variable on Voice, and this step leads to identification; while the final denotation ends up having the same definite description appearing as the argument of both Theme and Agent, this result has been arrived at through semantic binding, not by the same element saturating one function in its base position and one in its landing site. More broadly, since we do not begin from a conception of thematic roles as syntactic entities, compliance with the Theta Criterion is simply not at stake in the first place (Heim and Kratzer, 1998, 51). The view of thematic interpretation we assume – whereby thematic roles are simply neo-Davidsonian functions introduced at LF – in fact predicts different kinds of misalignments between syntactic structure and thematic interpretation that are not compatible with strict biuniqueness between syntactic arguments and thematic roles (see e.g. Legate 2014; Bruening 2013 on implicit agents in passives, and Wood 2015; Myler 2016 on so-called *delayed saturation*).

Our analysis of Turkish also differs in scope from that in Hornstein 2001. The theory Hornstein 2001 is intended as a general theory of referential dependency, covering not just cases of reflexivization without an obvious pronominal source, but also cases of antecedent-anaphor binding (where the anaphor is analyzed as the realization of a lower copy of the mobile antecedent) and of control. This theory further has cross-linguistic ambitions, intended as an account of construal across languages.

Our claims are more modest. We do not wish to claim that movement-based identification is the only strategy to build verbal reflexives cross-linguistically; put simply, we consider the Turkish case noteworthy precisely because it seems to clearly evidence a type of derivation at best underevidenced, and at worst counterevidenced, elsewhere. The derivation of English *shave* in (44) is a case in point. If thematic role identification in English proceeded as it does in Turkish, namely, by movement within the VoiceP, then we would expect the surface subject of such verbs in English to pass the same kinds of low origin diagnostics that its Turkish counterpart does. But this is not the case; for instance, English reflexive verbs do not license resultatives (see fn. 6, and cf. Reinhart and Siloni 2005 for more evidence in favor of a basic unergative syntax for English reflexive verbs).

This is not to say that the individual pieces of our analysis cannot be expected to be found in other languages; in the next section we offer some first directions for predictive avenues that our analysis opens up. But we do not claim that our analysis should be ported over wholesale to other structures that have been discussed under the label ‘verbal reflexive’. For different views on the extent to which the syntax of categories described under this label should be expected to be uniform crosslinguistically, see Kallulli and Roberts (2025); Reinhart and Siloni (2005); Marelj and Reuland (2016); Embick

(2004b); Alexiadou and Schäfer (2013); cf. Legate (2021) on the non-uniformity of ‘passives’.

Our discussion also explicitly prohibits us from extending the movement-based analysis to pronominal reflexives, again *pace* Hornstein (2001). If we were to assign the same derivation to verbal and pronominal reflexives in Turkish, the many differences between the two that we have highlighted would go unexplained; this specific concern is superimposed on the many independent concerns with positing movement as *the* way to relate reflexive pronouns and their antecedents, some familiar from parallel critiques of the movement theory of control (e.g., Landau 2003; Bobaljik and Landau 2009; Sigurðsson 2008; Wood 2012).

A final movement-based view of reflexivization worth comparing with our own is the analysis of Hebrew reflexive verbs in Kastner 2017. This analysis adjoins to nonactive Voice an agentive modifier $\sqrt{\text{ACTION}}$; the agent role contributed by this modifier is saturated by movement of the internal argument to a higher position, taken to be Spec,TP. The way the analysis is implemented requires a non-standard view of the interpretation of movement, one where the moving element is interpreted as a definite description in both its base position and landing site (see Kastner 2017, 14-15; Kastner does not directly diagnose the placement of the surface subject, but nonactive morphology is indeed suggestive of its low origin). We consider this view to be in need of further elaboration, since it does not capture the interpretation of movement chains elsewhere. Our analysis does not require a modifier of the $\sqrt{\text{ACTION}}$ type.

6 Conclusion

We have shown that the surface subject (and sole argument) of Turkish verbal reflexives occupies two positions within the VoiceP, patterning as an internal argument for some diagnostics and as an external argument for others, suggesting movement from the former to the latter position. We have taken it not to be an accident that we find movement between the position canonically associated with the theme role and that associated with the agent role; and that this type of movement is attested only in verbal reflexives, which involve the identification of these two roles, but not in other structures such as unaccusatives or passives. We have proposed a movement-triggering variant of the Reflexive Voice head, coupled with a view of its semantics that succeeds in deriving a reflexive interpretation from a single-argument syntax.

Throughout, we have argued for the correctness of this approach for the case of Turkish. Though we cannot exhaustively examine its cross-linguistic applicability, we conclude by outlining certain new avenues it opens up for cross-linguistic inquiry.

These predictions involve the thematic restrictions that characterize verbal reflexives. Cross-

linguistically, structures described as verbal reflexives tend to be agent-oriented (see Reuland 2018 for summary). An analysis that ties verbal reflexivization to the introduction of agents in syntax by means of Reflexive Voice derives such restrictions straightforwardly (Alexiadou, 2014; Spathas et al., 2015; McGinnis, 2022; Paparounas, 2023): as long as Voice heads more generally are taken to introduce Agents, the reflexive flavor will be capable of only deriving agent-oriented reflexives. We believe the more specific view offered here, where Reflexive Voice can be combined with movement, raises the possibility of explaining more finer-grained restrictions concerning the range of roles identified with the agent in Voice-based reflexivization.

On our analysis, Voice-based reflexivization is effected jointly by the free variable contributed by the Voice_{REFL} head, and by syntactic movement of the internal argument that yields binding of that free variable at LF. This view makes an interesting prediction: if to be reflexivized via Voice is to A-move in order to bind the free variable on Voice_{REFL}, then independent restrictions on A-movement will constrain the range of reflexivization possibilities. In particular, only elements independently capable of undergoing A-movement will be able to participate in Voice-based reflexivization. In particular, the account developed thus far raises two expectations:

- (45) a. In a given language $\mathcal{L}$ allowing Voice-reflexivization, if a DP α originating in a position canonically associated with a given thematic role θ can undergo A-movement to Spec,VoiceP, then θ can be identified with the role Agent, all things being equal.
b. If A-movement from a position canonically associated with a given thematic role θ is independently impossible in $\mathcal{L}$, then θ cannot be Voice-reflexivized with Agents in $\mathcal{L}$.

Here, we limit ourselves to two indicative initial looks into these expectations. Examining whether they are borne out more widely is one way to test the cross-linguistic applicability of our analysis.

A first data point is Turkish-internal. Key (2023, 2024) discusses two types of Turkish verbal reflexives differing as to which role is identified with the agent: figure (i.e. *theme*) or ground (an umbrella term for the roles *location*, *beneficiary* and *experiencer*). The sole argument in these reflexives is introduced in distinct projections: *vP* for figure reflexives and *ApplP* for ground reflexives. This bifurcation has the benefit of capturing the existence of transitive ground reflexives in Turkish, which involve a separate theme/figure argument separate from the ground, as in (46).

- (46) Davetsiz misafir-ler sandviç-ler-i tük-in-di-lar.
uninvited guest-PL sandwich-PL-ACC stuff-REFL-PST-PL
‘The uninvited guests stuffed themselves with sandwiches.’

Transitive ground reflexives can be incorporated into our analysis as in (47). This structure contains a separate figure DP, making for a transitive structure; the surface subject originates in the spec-

ifier of Appl, and movement to spec, Voice_{REFL}P derives identification of the agent with the ground.

- (47) [VoiceP [the uninvited guests] [APPLP [~~the uninvited guests~~] [vP sandwiches stuffed]]]
- ↑

The derived subject is the theme/figure argument of ordinary TVRs, but it is the ground argument if it is present, as in (46). On the theory proposed here, this follows from ordinary minimality restrictions on A-movement. The higher DP, the ground, will move to the specifier of Reflexive Voice if present; in its absence, the lower figure will instead. The movement trigger on Voice_{REFL} is indifferent as to the base position (and thus the base thematic role) of the lower element that will come to bind the free variable: whatever is closest moves, and what moves binds.

As a second point illustrating prediction (45b), consider Greek verbal reflexives formed with the reflexivizer *afto-*, which are obligatorily agent-oriented but not otherwise lexically restricted (Spathas et al., 2015; Alexiadou, 2014; Embick, 2004b; Paparounas, 2023). When ditransitive predicates are reflexivized in Greek, the theme is identified with the agent across a goal; notably, the goal is never identified with the agent (see Alexiadou 2014). This fact correlates with an independent property of Greek goal DPs: they are A-inert, never being promoted under passivization (but acting as defective interveners, see Anagnostopoulou 2003b). If the analysis advanced here for Turkish were extended to Greek, this fact would receive an automatic explanation: because Greek dative goals never A-move, they can neither be promoted under passivization nor be identified with the agent in verbal reflexivization. Note, though, that this explanation is promissory insofar as it presupposes independent evidence for movement-based reflexivization in Greek, a non-trivial question (see Paparounas 2024).

Our account then makes available a way of understanding thematic variation in verbal reflexivization by tying it to an independent factor, namely, language-particular constraints on A-movement. This avenue is worth exploring; we consider it a sign of progress that it is eminently falsifiable.

References

- Abels, Klaus. 2012. *Phases: An essay on cyclicity in syntax*. Berlin/Boston: Walter de Gruyter.
- Abusch, Dorit. 1989. Reflexives, Reference Shifters and Attitudes. In *Proceedings of the 8th West Coast Conference on Formal Linguistics*. Stanford: CSLI.
- Acartürk, Cengiz. 2005. Gradient characteristics of the unergative/unaccusative distinction in Turkish: An experimental investigation. Master's thesis, The Middle East Technical University.
- Acartürk, Cengiz, and Deniz Zeyrek. 2010. Unaccusative/unergative distinction in Turkish: A connectionist approach. In *Proceedings of the 8th Workshop on Asian Language Resources*, ed. by

- Virach Sornlertlamvanich Sarmad Hussain and Hammam Riza, 111–119. Beijing, China: Coling 2010 Organizing Committee.
- Ahn, Byron Thomas. 2015. Giving Reflexivity a Voice: Twin Reflexives in English. Doctoral Dissertation, University of California, Los Angeles.
- Akkuş, Faruk. 2021. (Implicit) Argument Introduction, Voice and Causatives. Doctoral Dissertation, University of Pennsylvania.
- Akkuş, Faruk. 2023. Turkish causatives embed a thematic VoiceP, and the Causee is an argument. In *Proceedings of the 53rd Annual Meeting of the North East Linguistic Society (NELS 53)*, ed. by Suet-Ying Lam and Satoru Ozaki, 1–14. GLSA Publications.
- Akkuş, Faruk, and Lefteris Paparounas. 2024. Turkish verbal reflexives and movement. In *Proceedings of the 54th Annual Meeting of the North East Linguistic Society (NELS 54)*, ed. by Shaunak Phadnis, Carla Spellerberg, and Brynne Wilkinson, volume 1, 21–30. Amherst, MA: GLSA Amherst.
- Alexiadou, Artemis. 2014. Roots in transitivity alternations: Afto-/auto-reflexives. In *The Syntax of Roots and the Roots of Syntax*, ed. by Artemis Alexiadou, Hagit Borer, and Florian Schäfer, 57–80. Oxford University Press.
- Alexiadou, Artemis, and Elena Anagnostopoulou. 2004. Voice morphology in the causative-inchoative Alternation: evidence for a non unified structural analysis of unaccusatives. In *The Unaccusativity Puzzle: Explorations of the Syntax-Lexicon Interface*, ed. by Artemis Alexiadou, Elena Anagnostopoulou, and Martin Everaert, 115—136. Oxford University Press.
- Alexiadou, Artemis, Elena Anagnostopoulou, and Florian Schäfer. 2006. The properties of anti-causatives crosslinguistically. In *Phases of interpretation*, ed. by Mara Frascarelli, volume 91, 187–211. Mouton de Gruyter.
- Alexiadou, Artemis, Elena Anagnostopoulou, and Florian Schäfer. 2015. *External arguments in transitivity alternations: A layering approach*. Oxford: Oxford University Press.
- Alexiadou, Artemis, and Edit Doron. 2012. The syntactic construction of two non-active Voices: Passive and middle. *Journal of Linguistics* 48:1–34.

- Alexiadou, Artemis, and Florian Schäfer. 2013. Towards a non-uniform analysis of naturally reflexive verbs. In *Proceedings of WCCFL 31*, ed. by Robert E. Santana-LaBarge. Somerville, MA: Cascadilla Proceedings Project.
- Altshuler, Daniel, Terence Parsons, and Roger Schwarzschild. 2019. *A course in semantics*. Cambridge, MA: MIT Press.
- Amato, Irene. 2024. Auxiliary selection in Italian restructuring: An insight into the size of the clause. *Syntax* 27:85–122.
- Anagnostopoulou, Elena. 2003a. Participles and Voice. In *Perfect Explorations*, ed. by Artemis Alexiadou, Monika Rathert, and Armin von Stechow, 1–36. De Gruyter.
- Anagnostopoulou, Elena. 2003b. *The syntax of ditransitives: Evidence from clitics*. Berlin/New York: Mouton de Gruyter.
- Arslan, Anıl. 2022. Türkiye Türkçesinde Dönüşlü Çatının Oluşum Yöntemleri. *Uluslararası Beşeri Bilimler ve Eğitim Dergisi (International Journal of Humanities and Education)* 8:226–240.
- Atlamaz, Ümit, and Balkız Öztürk. 2023. Reciprocals in Turkish. In *Languages: Theoretical Studies on Turkic Languages*, ed. by Jaklin Kornfilt, 158. MDPI.
- Baker, Mark. 1997. Thematic roles and syntactic structure. In *Elements of grammar*, ed. by Liliane Haegeman, 73–137. Dordrecht: Kluwer.
- Baker, Mark. 2022. On Agree without agreement as a source of reflexive voice constructions. *Glossa: a journal of general linguistics* 7.
- Barker, Chris. 2007. Parasitic scope. *Linguistics and Philosophy* 30:407–444.
- Bešlin, Maša. 2022. Revisiting Passive Participles: Category Status and Internal Structure. *Linguistic Inquiry* 1–66.
- Bhatt, Rajesh, and David Embick. 2004/2017. Causative derivations in Hindi-Urdu. *Journal of Indian Linguistics* 78:1-2:93–151.
- Biggs, Alison. 2019a. No lexical override in external argument interpretation. Talk given at GLOW 42.
- Biggs, Alison. 2019b. Objects in motion verb phrases. *Glossa: a journal of general linguistics* 4.

- Biktimir, Tuvana. 1986. Impersonal passives and the *-arak* construction in Turkish. In *Studies in Turkish linguistics*, ed. by Dan Isaac Slobin and Karl Zimmer, 53–75. Amsterdam/Philadelphia: Benjamins.
- Bobaljik, Jonathan David, and Idan Landau. 2009. Icelandic Control Is Not A-Movement: The Case from Case. *Linguistic Inquiry* 40:113–132.
- Boeckx, Cedric, Norbert Hornstein, and Jairo Nunes. 2008. Copy-reflexive and copy-control constructions: A movement analysis. *Linguistic Variation Yearbook* 8:61–100.
- Borer, Hagit. 2005. *Structuring Sense, Volume II: The normal course of events*. Oxford: Oxford University Press.
- Borer, Hagit. 2013. The syntactic domain of content. In *Language Acquisition and Language Disorders*, ed. by Misha Becker, John Grinstead, and Jason Rothman, volume 54, 205–248. Amsterdam: John Benjamins.
- Bouchard, Denis. 1982. On the content of empty categories. Doctoral Dissertation, MIT.
- Bresnan, Joan. 1982. The passive in lexical theory. In *The mental representation of grammatical relations*, ed. by Joan Bresnan, 3–86. Cambridge, MA: MIT Press.
- Bruening, Benjamin. 2006. The morphosyntax and semantics of verbal reciprocals. Ms., University of Delaware.
- Bruening, Benjamin. 2013. By-phrases in passives and nominals. *Syntax* 16:1–41.
- Bruening, Benjamin. 2021. Implicit arguments in English double object constructions. *Natural Language & Linguistic Theory* 39:1023–1085.
- Büring, Daniel. 2005. *Binding theory*. Cambridge Textbooks in Linguistics. Cambridge; New York: Cambridge University Press.
- Burzio, Luigi. 1981. Intransitive verbs and Italian auxiliaries. Doctoral Dissertation, MIT, Cambridge, MA.
- Burzio, Luigi. 1986. *Italian syntax: A Government and Binding approach*. Dordrecht: D. Reidel.

- Carlson, Greg N. 2006. The meaningful bounds of incorporation. In *Non-definiteness and plurality*, ed. by Svetlana Vogeleer and Liliane Tasmowski, volume 95, 35–50. John Benjamins Publishing Company.
- Çetinoğlu, Özlem, Miriam Butt, and Kemal Oflazer. 2008. Mono/bi-clausality of Turkish Causatives. In *Essays on Turkish Linguistics: Proceedings of the 14th International Conference on Turkish Linguistics*, ed. by Sıla Ay, Özgür Aydın, Iclal Ergenç, Seda Gökmen, Selçuk İşsever, and Dilek Peçenek, 43–53. Wiesbaden.
- Chierchia, Gennaro. 2004. A semantics for unaccusatives and its syntactic consequences. In *The unaccusativity puzzle: explorations of the syntax-lexicon interface*, ed. by Artemis Alexiadou, Elena Anagnostopoulou, and Martin Everaert, 22–59. Oxford: Oxford University Press.
- Chomsky, Noam. 1973. Conditions on transformations. In *A Festschrift for Morris Halle*, ed. by Stephen Anderson and Paul Kiparsky, 232–286. New York: Rinehart and Winston.
- Chomsky, Noam. 1981. *Lectures on government and binding*. Dordrecht: Foris Publications.
- Chomsky, Noam. 1993. A minimalist program for linguistic theory. In *The view from building 20: Essays in linguistics honor of Sylvain Bromberger*, ed. by Ken Hale and Samuel Keyser, 1–52. MIT Press.
- Chomsky, Noam. 1995. *The minimalist program*. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2008. On phases. In *Foundational issues in linguistic theory. essays in honor of Jean-Roger Vergnaud*, ed. by Robert Freidin, Carlos P Otero, and Maria Luisa Zubizarreta, 134–166. Cambridge, MA: MIT Press.
- Collins, Chris. 2024. *Principles of Argument Structure: A merge-based approach*. Number 87 in Linguistic Inquiry Monographs. Cambridge, MA: MIT Press.
- Dahl, Östen. 1973. On so-called ‘sloppy identity’. *Synthese* 81–112.
- Deal, Amy Rose. 2019. Raising to ergative: Remarks on applicatives of unaccusatives. *Linguistic Inquiry* 50:388–415.
- Dimitriadis, Alexis, and Martin Everaert. 2014. How many theta roles in a reflexive verb? *Acta Linguistica Hungarica* 61:247–269.

- Doron, Edit. 2003. Agency and voice: the semantics of the Semitic templates. *Natural Language Semantics* 11:1–67.
- Embick, David. 1997. Voice and the interfaces of syntax. Doctoral Dissertation, University of Pennsylvania.
- Embick, David. 2004a. On the structure of resultative participles in English. *Linguistic Inquiry* 35:355–392.
- Embick, David. 2004b. Unaccusative Syntax and Verbal Alternations. In *The unaccusativity puzzle: Explorations of the syntax-lexicon interface*, ed. by Artemis Alexiadou, Elena Anagnostopoulou, and Martin Everaert, 137–158. Oxford; New York: Oxford University Press.
- Embick, David. 2023. Smaller syntax for English stative passives: A first report. *Acta Linguistica Academica* 70:285–316.
- Embick, David, and Alec Marantz. 2008. Architecture and Blocking. *Linguistic Inquiry* 39:1–53.
- Fauconnier, Gilles. 1985. *Mental spaces*. Cambridge, MA: MIT Press.
- Fox, Danny. 1999. Reconstruction, Binding Theory, and the Interpretation of Chains. *Linguistic Inquiry* 30:157–196.
- Geniušienė, Emma. 1987. *The typology of reflexives*. Number 2 in Empirical Approaches to Language Typology. Berlin: De Gruyter.
- Georgi, Doreen. 2017. Patterns of movement reflexes as the result of the order of Merge and Agree. *Linguistic Inquiry* 48:585–626.
- Göksel, Asli. 1993. Levels of representation and argument structure in Turkish. Doctoral Dissertation, SOAS.
- Göksel, Asli, and Celia Kerslake. 2005. *Turkish: A comprehensive grammar*. London/New York: Routledge.
- Grimshaw, Jane. 1982. On the Lexical Representation of Romance Reflexive Clitics. In *The mental representation of grammatical relations*, ed. by Joan Bresnan, 87–148. Cambridge, MA: MIT Press.
- Grohmann, Kleanthes. 2003. *Prolific domains: On the anti-locality of movement dependencies*. Amsterdam/Philadelphia: John Benjamins.

- Gündoğdu, Songül. 2017. -(I)l / -(I)n morphology in Turkish: Implications for u-syncretism. In *4th Patras International Conference for Graduate Students in Linguistics (PICGL-4)*, 85–105.
- Gürer, Aslı. 2014. Adjectival participles in Turkish. *Lingua* 149:166–187.
- Göksel, Aslı, and Balkız Öztürk. 2019. *Conditions on prominent internal possessors in Turkish*, 163–195. Oxford University Press.
- Heim, Irene, and Angelika Kratzer. 1998. *Semantics in generative grammar*. Oxford: Blackwell Publishing.
- Hewett, Matthew. 2023. Types of Resumptive A-Dependencies. Doctoral Dissertation, University of Chicago.
- Holisky, Dee Ann. 1987. The case of the intransitive subject in Tsova-Tush (Batsbi). *Lingua* 71:102–132.
- Hopperdietzel, Jens. 2024. Themes to instruments: Applied arguments in Mandarin resultatives. paper presented at Handout from talk given at the Syntactic Theory at Rutgers colloquium at Rutgers University.
- Hornstein, Norbert. 1999. Movement and control. *Linguistic Inquiry* 30:69–96.
- Hornstein, Norbert. 2001. *Move! : A minimalist theory of construal*. Number 5 in Generative Syntax. Malden, MA/Oxford: Blackwell.
- Jackendoff, Ray. 1992. Mme. Tussaud Meets the Binding Theory. *Natural Language & Linguistic Theory* 10:1–31.
- Jo, Jinwoo. 2019. Encoding reflexivity: The syntax and semantics of natural reflexives. *Proceedings of the Linguistic Society of America* 4:2:1–15.
- Kallulli, Dalina, and Ian Roberts. 2025. On the relation between Voice and Case: An Agree-based approach. URL <https://ling.auf.net/lingbuzz/008426>, manuscript, University of Vienna and University of Cambridge.
- Kastner, Itamar. 2016. Form and meaning in the Hebrew verb. Doctoral Dissertation, NYU.
- Kastner, Itamar. 2017. Reflexive verbs in Hebrew: Deep unaccusativity meets lexical semantics. *Glossa: a journal of general linguistics* 2.

- Kaynak, Latife. 2012. Değerler Eğitimi Bağlamında Sevinç Çokum’un Hikayelerinin Analizi [The Analysis of Sevinç Çokum’s Stories in the context of Value Education]. Master’s thesis, Akdeniz University. URL <http://acikerisim.akdeniz.edu.tr/bitstream/handle/123456789/1945/?sequence=1&isAllowed=y>.
- Kayne, Richard S. 1975. *French syntax: The transformational cycle*. Number 6 in Current Studies in Linguistics. Cambridge, MA: MIT Press.
- Kayne, Richard S. 1988. Romance *se/si*. In *GLOW Newsletter* 20, 33. Budapest, Hungary.
- Kemmer, Susanne. 1993. *The middle voice*. Amsterdam/Philadelphia: John Benjamins.
- Key, Gregory. 2023. Figure and ground reflexives in Turkish. In *Proceedings of the Workshop on Turkic and Languages in Contact with Turkic 6*, ed. by Songül Gündoğdu, Sahar Taghipour, and Andrew Peters, 1–15. Linguistic Society of America.
- Key, Gregory. 2024. Voice in Turkish: Re-thinking *u*-syncretism. *Natural Language & Linguistic Theory*.
- Knecht, Laura Ellen. 1985. Subject and Object in Turkish. Doctoral Dissertation, MIT.
- Kornfilt, Jaklin. 1997. *Turkish*. London/New York: Routledge.
- Kornfilt, Jaklin. 2001. Local and long-distance reflexives in Turkish. In *Long-distance reflexives*, ed. by Peter Cole, Gabriella Hermon, and C.-T. James Huang, 197–226. San Diego, CA: Academic Press.
- Kratzer, Angelika. 2009. Making a pronoun: Fake indexicals as windows into the properties of pronouns. *Linguistic Inquiry* 40:187–237.
- Krejci, Bonnie. 2020. Syntactic and semantic perspectives on first conjunct agreement in Russian. Doctoral Dissertation, Stanford University.
- Labelle, Marie. 2008. The French reflexive and reciprocal *se*. *Natural Language & Linguistic Theory* 26:833–876.
- Landau, Idan. 2003. Movement out of control. *Linguistic Inquiry* 34:471–498.
- Lee, Soo-Hwan. 2024. Introducing arguments and case. Doctoral Dissertation, New York University.

- Legate, Julie Anne. 2014. *Voice and v: Lessons from Acehnese*. Cambridge, MA: MIT Press.
- Legate, Julie Anne. 2021. Noncanonical Passives: A Typology of Voices in an Impoverished Universal Grammar. *Annual Review of Linguistics* 7:157–176.
- Legate, Julie Anne, Faruk Akkuş, Milena Šereikaitė, and Don Ringe. 2020. On passives of passives. *Language* 96:771–818.
- Levin, Beth, and Malka Rappaport Hovav. 1995. *Unaccusativity: At the syntax-lexical semantics interface*. Cambridge, MA: MIT Press.
- Lewis, Geoffrey. 1967. *Turkish grammar*. Oxford: Oxford University Press.
- Lidz, Jeffrey. 1996. Dimensions of reflexivity. Doctoral Dissertation, University of Delaware.
- Lidz, Jeffrey. 2001a. The argument structure of verbal reflexives. *Natural Language & Linguistic Theory* 19:311–353.
- Lidz, Jeffrey. 2001b. Condition R. *Linguistic Inquiry* 32:123–140.
- Marantz, Alec. 1984. *On the nature of grammatical relations*. Cambridge, MA: MIT Press.
- Marantz, Alec. 1997. No escape from syntax: Don't try morphological analysis in the privacy of your own lexicon. In *Proceedings of the 21st Annual Penn Linguistics Colloquium*, 201–225. Philadelphia: Penn Working Papers in Linguistics.
- Marelj, Marijana, and Eric Reuland. 2016. Clitics and Reflexives: Reducing the Lexicon-Syntax Parameter. In *Concepts, Syntax, and Their Interface: The Theta System*, ed. by Martin Everaert, Marijana Marelj, and Eric Reuland, number 72 in Linguistic Inquiry Monographs, 175–252. MIT Press.
- Martí, Luisa. 2015. Grammar versus Pragmatics: Carving Nature at the Joints. *Mind & Language* 30:437–473.
- McCloskey, Jim. 1997. Subjecthood and subject positions. In *Elements of grammar*, ed. by Liliane Haegeman, 197–235. Kluwer Academic Publishers.
- McGinnis, Martha. 2004. Lethal Ambiguity. *Linguistic Inquiry* 35:47–95.

- McGinnis, Martha. 2022. Reflexive clitics are verbal, not pronominal. *Canadian Journal of Linguistics/Revue canadienne de linguistique* 67:328–352.
- McIntyre, Andrew. 2013. Adjectival passives and adjectival participles in English. In *Non-canonical passives*, ed. by Artemis Alexiadou and Florian Schäfer, volume 205, 21–42. John Benjamins Publishing Company.
- Merchant, Jason. 2019. Roots don't select, categorial heads do: Lexical-selection of PPs may vary by category. *The Linguistic Review* 36:325–341.
- Müller, Gereon. 2010. On deriving CED effects from the PIC. *Linguistic Inquiry* 41:35–82.
- Myler, Neil. 2016. *Building and interpreting possession sentences*. Cambridge, MA: MIT Press.
- Nakipoğlu-Demiralp, Mine. 1998. Split intransitivity and the syntax-semantics interface in Turkish. Doctoral Dissertation, University of Minnesota.
- Nakipoğlu-Demiralp, Mine. 2001. The referential properties of the implicit arguments of impersonal passive constructions. In *The verb in Turkish*, ed. by Eser Erguvanlı Taylan, 129–150. Amsterdam: John Benjamins.
- Nakipoğlu-Demiralp, Mine. 2002. Türkçe'deki ayrık geçişsiz eylemlerin olay yapısal bir incelemesi [An event-structural investigation of split-intransitivity in Turkish]. *Dilbilim Araştırmaları* 2002 1–16.
- Neu, Eva. 2026. Direct causatives of unergatives: A view from variable unaccusativity. *Natural Language & Linguistic Theory* 44:5.
- Oikonomou, Despina, and Artemis Alexiadou. 2022. Voice Syncretism Crosslinguistically: The View from Minimalism. *Philosophies* 7:19.
- Ordóñez, Francisco. 2021. On the Nature of the Impersonal SE: Why Italian is not like Catalan and Spanish. In *Unraveling the complexity of SE*, ed. by Grant Armstrong and Jonathan E. MacDonald, 137–160. Dordrecht: Springer.
- Özkaragöz, Inci. 1980. Evidence from Turkish for the unaccusative hypothesis. *Proceedings of the Sixth Annual Meeting of the Berkeley Linguistics Society* 411–422.

- Papangeli, Dimitra. 2004. The morpho-syntax of argument realization: Greek argument structure and the Lexicon-Syntax interface. phdthesis, LOT.
- Paparounas, Lefteris. 2023. Voice from Syntax to Syncretism. Doctoral Dissertation, University of Pennsylvania.
- Paparounas, Lefteris. 2024. Reflexivity and External Argument Introduction: Insights from Greek. URL <https://ling.auf.net/lingbuzz/007537>, ms, Université du Québec à Montréal.
- Paparounas, Lefteris, and Faruk Akkuş. 2024. Anaphora and agreement in the Turkish DP: Delimiting binding-through-Agree. *Natural Language & Linguistic Theory* 42:633–700.
- Parsons, Terence. 1990. *Events in the semantics of English: A study in subatomic semantics*. Number 19 in Current Studies in Linguistics. Cambridge, MA: MIT Press.
- Perlmutter, David. 1978. Impersonal passives and the Unaccusative Hypothesis. *Papers from the Annual Meeting of the Berkeley Linguistic Society* 4:157–189.
- Pesetsky, David. 1995. *Zero syntax*. Cambridge, MA: MIT Press.
- Poole, Ethan. 2024. (Im)possible Traces. *Linguistic Inquiry* 55:287–326.
- Pylkkänen, Liina. 2008. *Introducing arguments*. Cambridge, MA: MIT Press.
- Raghotham, Sreekar. 2022. UTAH, The Criterion Theta, and how they fare against Telugu data. URL <https://www.crissp.be/wp-content/uploads/2022/10/Raghotham.pdf>, handout given at the 15th Brussels Conference on Generative Linguistics (BCGL 15): Argument structure, theta-roles, and their realization.
- Raghotham, Sreekar. 2024. Lessons from the wax museum. In *Proceedings of the Fifty-ninth Annual Meeting of the Chicago Linguistic Society*, ed. by Kutay Serova and M. K. Snigaroff.
- Reinhart, Tanya. 1983. *Anaphora and semantic interpretation*. London: Croom Helm.
- Reinhart, Tanya. 2002. The Theta System – an overview. *Theoretical Linguistics* 28:229–290.
- Reinhart, Tanya. 2006. *Interface Strategies: Optimal and Costly Computations*. Cambridge, MA: The MIT Press.

- Reinhart, Tanya. 2016. The Theta System: Syntactic Realization of Verbal Concepts. In *Concepts, Syntax, and Their Interface*, ed. by Martin Everaert and Marijana Marelj, 1–112. MIT Press.
- Reinhart, Tanya, and Eric Reuland. 1993. Reflexivity. *Linguistic Inquiry* 24:657–720.
- Reinhart, Tanya, and Tal Siloni. 2004. Against an unaccusative analysis of reflexives. In *The unaccusativity puzzle: Explorations of the syntax-lexicon interface*, ed. by Artemis Alexiadou, Elena Anagnostopoulou, and Martin Everaert, number 5 in Oxford Studies in Theoretical Linguistics, 159–180. Oxford: Oxford University Press.
- Reinhart, Tanya, and Tal Siloni. 2005. The Lexicon-Syntax Parameter: Reflexivization and Other Arity Operations. *Linguistic Inquiry* 36:389–436.
- Reuland, Eric. 2001. Primitives of Binding. *Linguistic Inquiry* 32:439–492.
- Reuland, Eric. 2011. *Anaphora and language design*. Cambridge, MA: MIT Press.
- Reuland, Eric. 2018. Reflexives and Reflexivity. *Annual Review of Linguistics* 4:81–107.
- Reuland, Eric, and Yoad Winter. 2009. Binding without Identity: Towards a Unified Semantics for Bound and Exempt Anaphors. In *Anaphora Processing and Applications*, ed. by Sobha Lalitha Devi, António Branco, and Ruslan Mitkov, volume 5847, 69–79. Berlin: Springer.
- Safir, Ken. 2004. *The Syntax of Anaphora*. Oxford: Oxford University Press.
- Şahin, Müşerref. 2019. Selçuklular Devri Siyasetnamelerine Göre Devlet Adamının Niteliği. MA thesis, Necmettin Erbakan University.
- Sauerland, Uli. 2004. The Interpretation of Traces. *Natural Language Semantics* 12:63–127.
- Sauerland, Uli. 2013. Presuppositions and the alternative tier. In *Semantics and Linguistic Theory*, 156–173.
- Schäfer, Florian. 2008. *The syntax of (anti-) causatives: External arguments in change-of-state contexts*. Amsterdam/Philadelphia: John Benjamins.
- Schäfer, Florian. 2012. The passive of reflexive verbs and its implications for theories of binding and case. *The Journal of Comparative Germanic Linguistics* 15:213–268.

- Sells, Peter, Annie Zaenen, and Draga Zec. 1987. Reflexivization variation: Relations between syntax, semantics and lexical structure. In *Working papers in grammatical theory and discourse structure*, ed. by Masayo Iida, Stephen Wechsler, and Draga Zec, 169–23. Stanford, CA: CSLI.
- Sigurðsson, Halldór Ármann. 2008. The case of PRO. *Natural Language & Linguistic Theory* 26:403–450.
- Siloni, Tal. 2012. Reciprocal verbs and symmetry. *Natural Language & Linguistic Theory* 30:261–320.
- Simpson, Jane. 1983. Resultatives. In *Papers in Lexical-functional Grammar*, ed. by Lori Levin, Malka Rappaport, and Annie Zaenen, volume 6, 143–157. Bloomington, IN: Indiana University Linguistics Club.
- Sorace, Antonella. 2000. Gradients in auxiliary selection with intransitive verbs. *Language* 76:859–890.
- Spathas, Giorgos, Artemis Alexiadou, and Florian Schäfer. 2015. Middle Voice and reflexive interpretations: afto-prefixation in Greek. *Natural Language & Linguistic Theory* 33:1293–1350.
- Sportiche, Dominique. 2005. *Partitions and Atoms of Clause Structure*. London: Routledge.
- Sportiche, Dominique. 2014. Assessing Unaccusativity and Reflexivity: Using Focus Alternatives to Decide What Gets Which θ -Role. *Linguistic Inquiry* 45:305–321.
- Sundaresan, Sandhya. 2012. Context and (Co)reference in the syntax and its interfaces. Doctoral Dissertation, University of Tromsø.
- Turgay, Tacettin. 2013. Resultative Constructions in Turkish. Master's thesis, Boğaziçi University.
- Wechsler, Stephen. 1997. Resultative predicates and control. In *Texas Linguistics Forum*, volume 38, 307–321. University of Texas at Austin.
- Wehrli, Eric. 1986. On some properties of French clitic *se*. In *The grammar of pronominal clitics*, ed. by Hagit Borer, 263–283. New York: Academic Press.
- Wood, Jim. 2012. Against the Movement Theory of Control: Another Argument from Icelandic. *Linguistic Inquiry* 43:322–330.

- Wood, Jim. 2014. Reflexive -st verbs in Icelandic. *Natural Language & Linguistic Theory* 32:1387–1425.
- Wood, Jim. 2015. *Icelandic morphosyntax and argument structure*. Cham: Springer.
- Zec, Draga. 1985. Objects in Serbo-Croatian. *Annual Meeting of the Berkeley Linguistics Society* 358–371.
- Zimmer, Karl E. 1976. Some constraints on Turkish causativization. In *The grammar of causative constructions*, ed. by M. Shibatani, 399–412. New York: Academic Press.

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Notes

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1. The characteristic syntax of TVRs has allowed us to reliably exclude from this large class verbs that bear non-active morphology but are not syntactically reflexive (e.g. change-of-state unaccusatives and different kinds of motion verbs).
2. Turkish judgments originate from the native speaker co-author; we provide examples attested in online sources and published studies where relevant. Regardless of their source, all examples have been confirmed with a approximately 80 Turkish speakers across multiple academic venues. For the data in section 4.4, we initially conducted an informal social media questionnaire with 9 speakers whose judgments have since been confirmed by additional speakers.

3. Note that the availability of resultatives with TVRs cannot be taken to follow merely from the presence of a theme entailment with TVRs, lest we fail to make the right cut between transitives, passives, unaccusatives and TVRs on the one hand, and examples like (14b) on the other. The resultatives-diagnose-theme-entailments view would also fail to make sense of minimal differences between Turkish and other languages. Consider English and Sason Arabic (data from the native speaker co-author). Reflexive verbs in these languages presumably involve theme entailments just as much as their Turkish counterparts do, yet resultatives are impossible:

- | | |
|----------------------------|--|
| (i) John shaved (*smooth). | (ii) <i>Nobody else washed Kemal. Instead, on his own...</i>
#Kemal in-xasal nazzif.
Kemal REFL-wash.PST.3M clean
#‘Kemal washed clean.’ (Sason Arabic) |
|----------------------------|--|

Whereas the presence/absence of theme entailments does not suffice to explain these differences between Turkish and English/Sason Arabic, the different syntaxes of reflexive verbs in these languages do. Tellingly, independent diagnostics confirm that English *shave*-type verbs have the syntax of unergatives (Reinhart and Siloni, 2004, 2005), and Sason Arabic reflexives do, too. For apparent counterexamples to the direct object restriction, see Biggs 2019a,b on Wechsler 1997.

4. In the interest of space, we leave out discussion of a further convergent diagnostic, namely long object movement. Causativization, deployed as an intransitivity diagnostic in section 2.2, in fact constitutes a further external argumenthood test; see discussion near the end of section 4.4.

5. Romance languages show similar effects, inviting identical analyses (Ordóñez, 2021, 155).

6. Most of the baseline generalizations on *-ArAk* are distributed across various studies; the behavior of TVRs in these gerunds was first noted in Akkuş and Paparounas 2024, replicated here. In this paper we only provide examples with TVRs in the interest of space.

7. An anonymous reviewer suggests examining *theme unergatives* (in the sense of Reinhart 2002; Levin and Rappaport Hovav 1995 and related work) as a way of dissociating thematic role from syntactic position, and thereby confirming that our positional diagnostics are sensitive to the former and not simply the latter (cf. footnote 6). In Turkish, such predicates (e.g. *kokuşmak* ‘stink’, *parıldamak* ‘sparkle’, *ciyaklamak* ‘squeal’) pattern with regular unergatives for every conceivable diagnostic (e.g. they freely form episodic impersonals; they never form adjectival participles; they pattern like other unergatives in *-ArAk* gerundives). If the sole argument of these verbs is a theme, then this data confirms that our diagnostics track syntactic position rather than thematic role. We thank the reviewer for raising this point.

8. A selection-free alternative to the issue would posit that the grammar in fact generates all possi-

ble combinations, with aberrant ones deemed impossible only at the level of world knowledge (e.g. [Marantz, 1997](#); [Borer, 2005](#)). Since we have argued that encyclopedic content alone (either at the level of lexical semantics or at that of world knowledge) does not always predict which Root will and will not form a TVR, this route is not available to us.

9. Situations like this are by no means uncommon. In examples like *Every student/Mary seems to herself to have passed the exam*, the embedded subject raises to matrix grammatical subject in a way that allows it to bind both its trace and the index on the reflexive pronoun located in the matrix clause (see e.g. [Reinhart 2006](#), 165ff, [Reuland 2011](#), 348). Note that quantified subjects of TVRs will receive the same analysis that quantified DPs receive elsewhere: lambda abstraction at the final landing site on a generalized quantifier analysis, or a higher type assigned to quantified expressions.

10. *Modulo* broader architectural consequences not specific to our analysis, our account could in principle be translated into a framework where indices and variables are assigned a narrow-syntactic life (see e.g. [Bruening 2021](#) for one such approach to certain implicit complements).

11. Not all verbal reflexives syncretize with passives in Turkish ([Kornfilt, 1997](#); [Göksel, 1993](#); [Key, 2023, 2024](#)). As Key notes, it is *ground* reflexives that are realized with a dedicated exponent not shared with the passive, see section 6.